FIITJEE

ALL INDIA TEST SERIES

PART TEST – II

JEE (Main)-2020

TEST DATE: 16-11-2019

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A

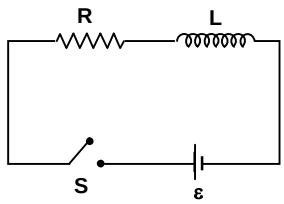
(One Options Correct Type)

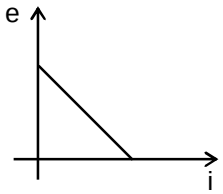
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

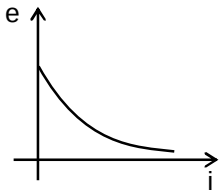
1. A sphere of radius 3.5 cm and mass 250 gm floats in a liquid at 20°C. As the temperature is raised, the sphere just begins to sink at a temperature of 35°C. If the density of liquid at 0°C is 1.527 gm/cm^3, find the coefficient of cubical expansion of the liquid. Neglect expansion of the sphere.
   (A) 1.30×10^{-3} C^{-1}
   (B) 5.61×10^{-3} C^{-1}
   (C) 2.75×10^{-3} C^{-1}
   (D) 8.68×10^{-5} C^{-1}

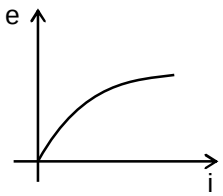
2. In a certain region of outer space there are only 100 molecules per cm^3 on an average. The temperature there is about 3K. What is the average pressure of this very dilute gas?
   (A) 4.14×10^{-15} Pa
   (B) 1.14×10^{-15} Pa
   (C) 8.28×10^{-14} Pa
   (D) 9.50×10^{-14} Pa

3. In an L-R circuit connected to a battery of constant emf ε, switch S is closed at time t = 0. If e denotes the induced emf across inductor and i the current in the circuit at any time t. Then which of the following graphs shows the variation of e with i?

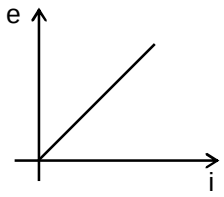


(A)


(B)


(C)


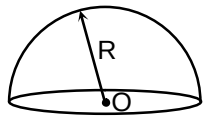
(D)



4. A parallel beam of nitrogen molecules moving with velocity v = 400 m/s impinges on a wall at an angle $\theta = 30°$ to its normal. The concentration of molecule in the beam $n_0 = 0.9 \times 10^{19}$ cm^{-3}. Find the pressure exerted by the beam on the wall assuming the collisions of molecules with the wall are perfectly elastic.
(A) 10^6 Pa
(B) 10^7 Pa
(C) 10^4 Pa
(D) 10^5 Pa

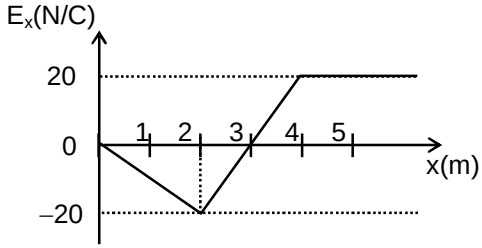
5. A beam of fast moving electrons having cross section area A falls normally on flat surface. The electrons are absorbed by the surface and the average pressure exerted by the electrons on this surface is found to be P. If the electrons are moving with a speed v, then the effective current through any cross section of the electrons beam is
(A) $\frac{APe}{(mv)}$
(B) $\frac{APe}{(mv^2)}$
(C) $\frac{2APe}{(mv)}$
(D) $\frac{APm}{(ev)}$

6. Charge Q coulombs is uniformly distributed through out the volume of a solid hemisphere of radius R meters. Then find the potential at centre O of the hemisphere in volts.



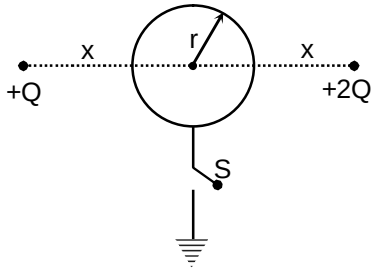
(A) $\frac{1}{4\pi\varepsilon_0}\frac{3Q}{2R}$
(B) $\frac{1}{4\pi\varepsilon_0}\frac{3Q}{4R}$
(C) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{4R}$
(D) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{8R}$

7. A graph of the X component of the electric field as a function of x in a region of space is shown. The Y and Z components of the electric field are zero in this region. If the electric potential is 10V at the origin, then find the potential at x = 2.0 m.



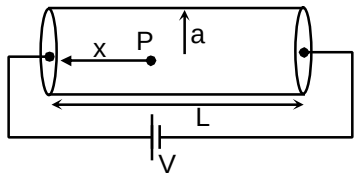
(A) 10 V
(B) 40 V
(C) −10 V
(D) 30 V

8. Two particles having positive charges +Q and +2Q are fixed at equal distance x from the centre of a conducting sphere having zero net charge and radius r as shown. Initially the switch S is open. After the switch S is closed, then find the net charge following out of sphere.



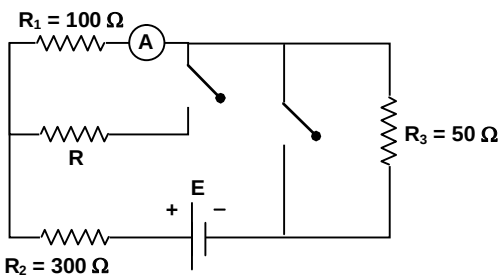
(A) $\frac{Qr}{x}$
(B) $-\frac{Qr}{x}$
(C) $\frac{3Qr}{x}$
(D) $-\frac{3Qr}{x}$

9. A cylindrical solid rod of length L and radius 'a' is connected across a source of emf V and negligible internal resistance as shown in the figure. The resistivity of the rod at point P at a distance 'x' from left end is given by $\rho = bx$ (where b is a positive constant). Find the electric field at point P.



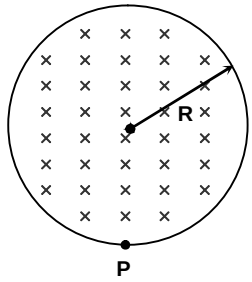
(A) $\frac{2V}{L^2}x$
(B) $\frac{V}{L^2}x$
(C) $\frac{4V}{L^2}x$
(D) $\frac{8V}{L^2}x$

10. In the circuit shown, the reading of the ammeter (ideal) is the same with both switches open as with both closed. Find the value of resistance R in Ω.



(A) 600
(B) 800
(C) 400
(D) 300

11. A uniform magnetic field of induction B is confined to cylindrical region of radius R. The magnetic field is increasing at a constant rate of $\frac{dB}{dt}(TS^{-1})$. An electron of charge –e, placed at the point P on the periphery of the field experiences an acceleration.



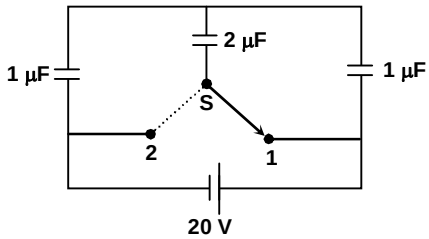
(A) $\frac{1}{2}\frac{eR}{m}\frac{dB}{dt}$ towards left

(B) $\frac{1}{2}\frac{eR}{m}\frac{dB}{dt}$ towards right

(C) $\frac{eR}{m}\frac{dB}{dt}$ towards left

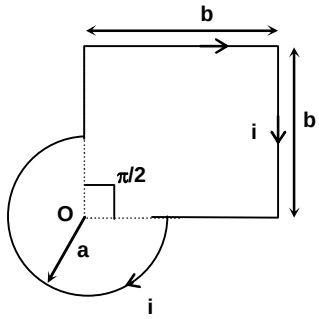
(D) zero

12. The amount of heat liberated when the switch S is shifted from position 1 to 2 is



(A) 180 μJ
(B) 200 μJ
(C) 220 μJ
(D) 210 μJ

13. Find the magnitude of magnetic field at O (centre of the circular part) of the current carrying coil as shown.



(A) $\frac{\mu_0 i}{4\pi}\left(\frac{3\pi}{a}+\frac{\sqrt{2}}{b}\right)$

(B) $\frac{\mu_0 i}{2\pi}\left(\frac{3\pi}{2a}+\frac{\sqrt{2}}{b}\right)$

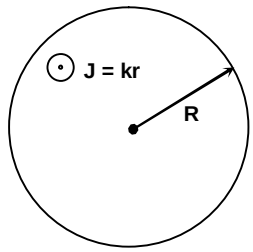
(C) $\frac{\mu_0 i}{2\pi}\left(\frac{\pi}{3a}+\frac{3}{\sqrt{2}b}\right)$

(D) $\frac{\mu_0 i}{4\pi}\left(\frac{3\pi}{2a}+\frac{\sqrt{2}}{b}\right)$

14. If a charged particle goes un accelerated in a region containing electric and magnetic fields, then which of the following is true?
    (A) $\vec{E}$ must be parallel to $\vec{B}$
    (B) $\vec{V}$ must be perpendicular to $\vec{E}$.
    (C) $\vec{E}$ must be equal to $\vec{B}$
    (D) $\vec{V}$ must be perpendicular to $\vec{B}$.

15. A nonconducting spherical shell of radius 'R' and uniformly charged on its surface with charge 'Q' is rotating about its diametrical axis with a constant angular velocity 'ω'. Find the magnetic moment of the sphere.
    (A) $\frac{QR^2\omega}{2}$
    (B) $\frac{QR^2\omega}{3}$
    (C) $\frac{QR^2\omega}{4}$
    (D) none of these

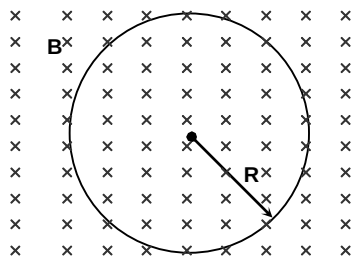
16. The current density (current per unit area) in a solid cylinder of radius R varies with distance r from its axis as J = kr, where k is a constant. Find the magnetic field at a point P where r > R.



    (A) $\frac{\mu_0 kR^2}{r}$
    (B) $\frac{\mu_0 kR^3}{r}$
    (C) $\frac{\mu_0 kR^3}{2r}$
    (D) $\frac{\mu_0 kR^3}{3r}$

17. An equilateral triangular loop of wire of side ℓ carries a current i. The magnetic field produced at the circum centre of the loop is
    (A) $\frac{\mu_0}{4\pi}\frac{3\sqrt{3}i}{\ell}$
    (B) $\frac{\mu_0}{4\pi}\frac{9i}{\ell}$
    (C) $\frac{\mu_0}{4\pi}\frac{18i}{\ell}$
    (D) $\frac{\mu_0}{4\pi}\frac{6i}{\ell}$

18. A conducting loop of radius R is present in a uniform magnetic field B perpendicular to the plane of the ring. If radius R varies as a function of time 't' as $R = R_0 + t$, the emf induced in the loop is



(A) $4\pi(R_0 + t)B$
(B) $\pi(R_0 + t)B$ clockwise
(C) $2\pi(R_0 + t)B$ anticlockwise
(D) zero

19. The magnetic flux through a conducting loop with resistance R varies during interval of time T as $\phi = at(T - t)$. The heat generated during this time, neglecting the inductance of loop will be

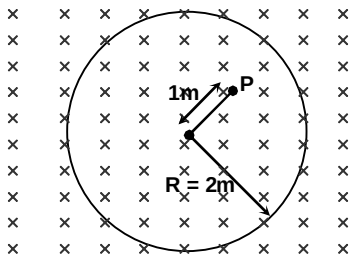
(A) $\frac{a^2T^3}{3R}$

(B) $\frac{a^2T^2}{3R}$

(C) $\frac{a^2T}{3R}$

(D) $\frac{a^2T^3}{R}$

20. The magnetic field in a certain cylindrical region is changing with time according to the law $B = (16 - 4t^2)$ Tesla. The induced electric field at point P at time t = 2sec.

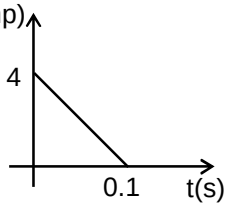


(A) 8 Volt/m
(B) 6 Volt/m
(C) 4 Volt/m
(D) none of these

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. When magnetic flux through a coil is changed, the variation of induced current in the coil with time is as shown in graph. If resistance of coil is 10Ω, then the total change in flux (in Weber) of coil will be

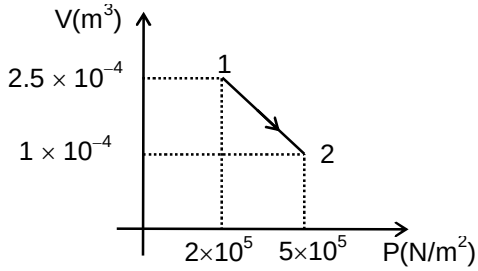


22. The potential at a point P due to a point electric dipole is 1.8×0^5 V. If P is at a distance of 50 cm from the centre O of the dipole and if vector $\overrightarrow{OP}$ makes an angle 60° with the positive side of the axial line of the dipole, the dipole moment of the dipole is 10^{-n} C-m. Find the value of n.

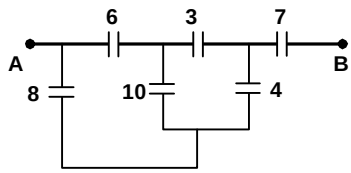
SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. A gas is taken from State-1 to State-2 along the path shown in the figure. If 70 cal of heat is rejected by the gas in the process. Find the magnitude of change in internal energy (in Joule) of the system.



24. In the circuit diagram shown, all the capacitors are in μF. The equivalent capacitance between points A and B is (in μF)



25. 15 gm of nitrogen is enclosed in a container of constant volume at temperature T = 300 K. Find the amount of heat (in Joule) required to double the root mean square velocity of these molecules.

Chemistry

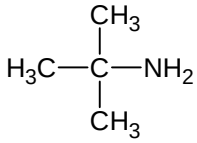
PART – II

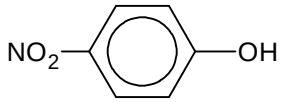
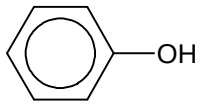
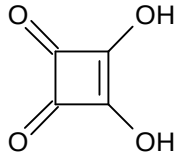
SECTION – A

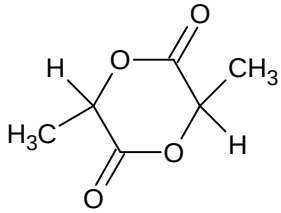
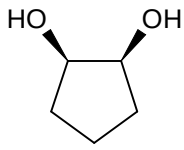
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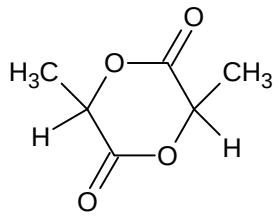
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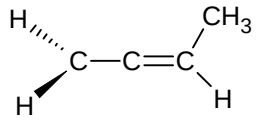
26. What is the degree of unsaturation of the compound formed by the reaction of formaldehyde with excess of ammonia?
(A) 6
(B) 4
(C) 3
(D) 5

27. Which of the amines can not be prepared by Gabriel's phthalimide synthesis?
(A) $CH_3 - CH_2 - CH_2 - NH_2$
(B) $CH_3 - CH_2 - NH_2$
(C) $CH_3 - NH_2$
(D) 

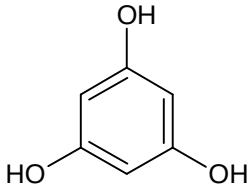
28. Which of the following acid is the strongest acid?
(A) 
(B) 
(C) $CH_3 - CH_2COOH$
(D) 

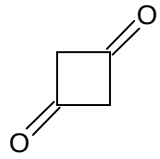
29. Which of the following molecule is chiral?
(A) 
(B) 

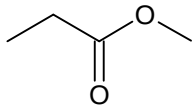
(C) 

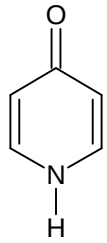
(D) 

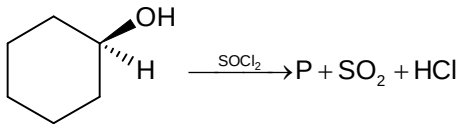
30. Which of the compound has most stable enol form?

(A) 

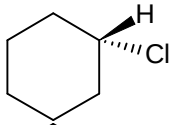
(B) 

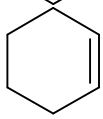
(C) 

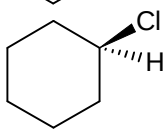
(D) 

31. 

The structure of P is:

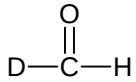
(A) 

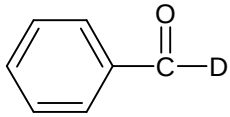
(B) 

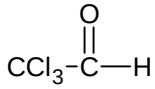
(C) 

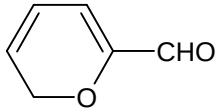
(D) Both A and C are in equal ratio

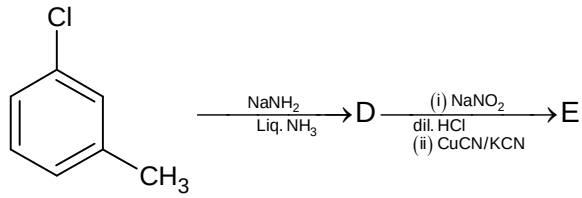
32. Which of the following compounds does not undergo Cannizzaro reaction?

(A) 

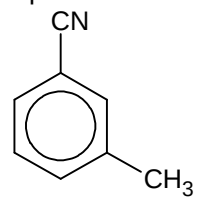
(B) 

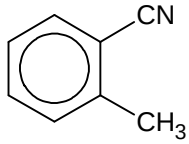
(C) 

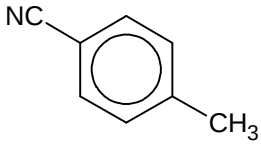
(D) 

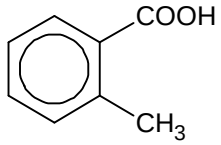
33. 

Find out the product E.

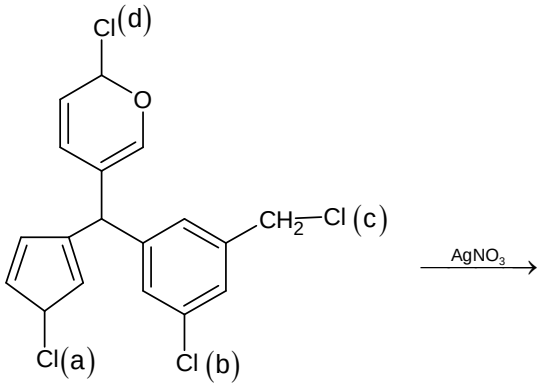
(A) 

(B) 

(C) 

(D) 

34.



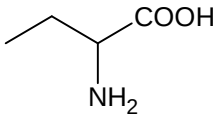
Which of the Cl is removed in fastest rate?

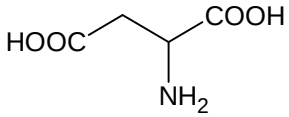
(A) Cl (a)
(B) Cl (b)
(C) Cl (c)
(D) Cl (d)

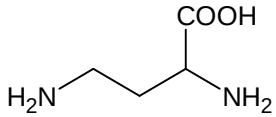
35. D-glucose and D-fructose can be distinguished by

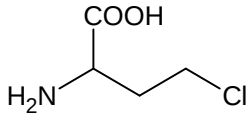
(A) Fehling's Test
(B) Tollen's Test
(C) Br_2/H_2O
(D) Osazone formation

36. Which of the amino acids is a basic amino acid?

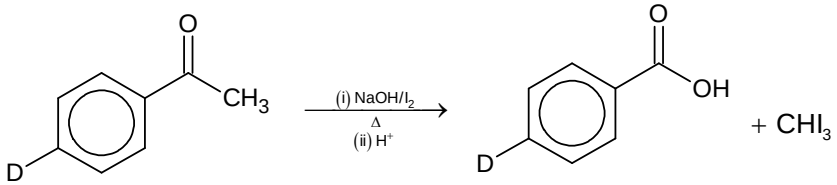
(A) 

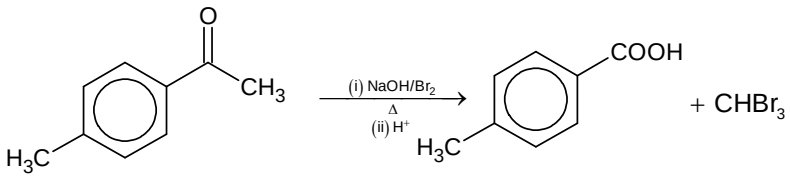
(B) 

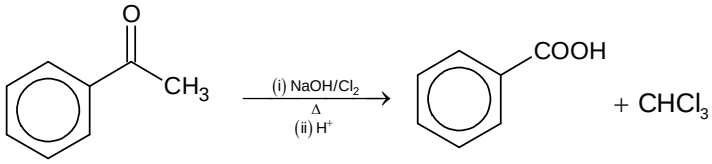
(C) 

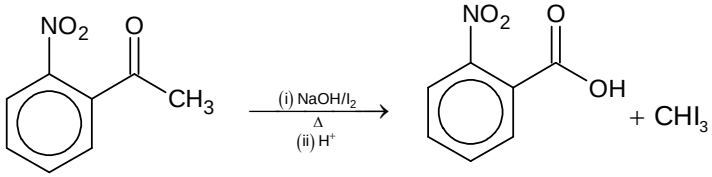
(D) 

37. Which of the reaction takes place in fastest rate?

(A) 

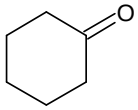
(B) 

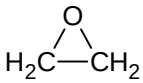
(C) 

(D) 

38. Which of the following compound produces primary alcohol on reaction with CH_3MgBr?

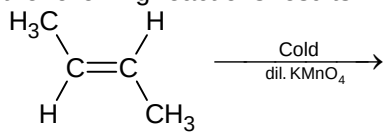
(A) $CH_3 - CH = O$

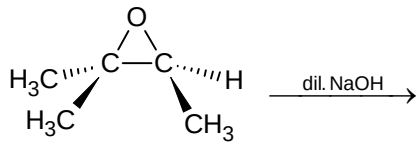
(B) 

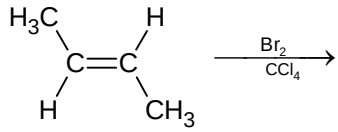
(C) 

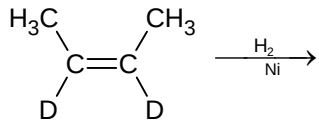
(D) $CH_3 - C \equiv N$

39. Which of the following reactions results meso compound as the product?

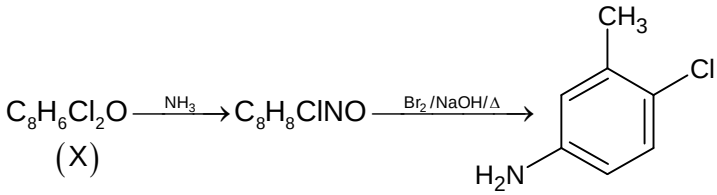
(A) 

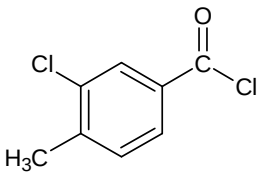
(B) 

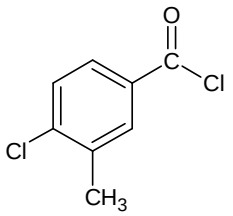
(C) 

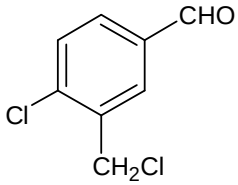
(D) 

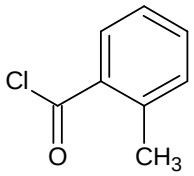
40. Identify 'X' in the following sequence of reaction



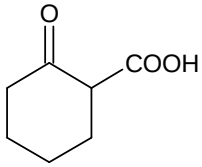
(A) 

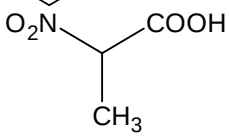
(B) 

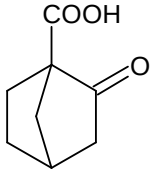
(C) 

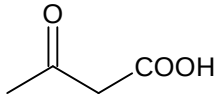
(D) 

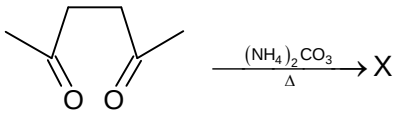
41. Which of the following compounds is most difficult to decarboxylate?

(A) 

(B) 

(C) 

(D) 

42. 

Which of the following statement is INCORRECT?

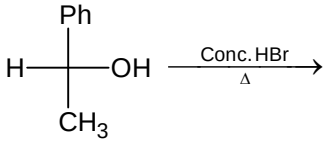
(A) The product X has a five membered ring.

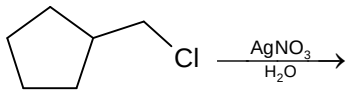
(B) The product X is aromatic.

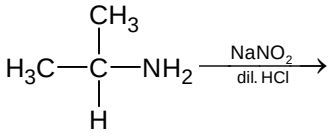
(C) The degree of unsaturation of the product X is 3.

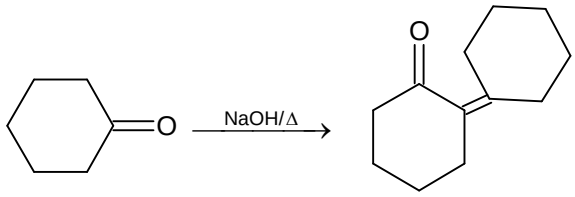
(D) The product X has a six membered ring.

43. Which of the following reactions does not proceed through the formation of a carbocation?

(A) 

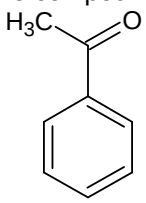
(B) 

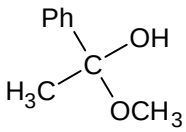
(C) 

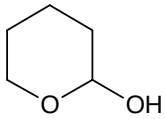
(D) 

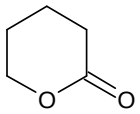
(2 eq.)

44. Which of the compounds can respond in Fehling's test?

(A) 

(B) 

(C) 

(D) 

45. Which of the following reaction leads to the Hoffmann's product?

(A) $CH_3 - CH_2 - CH_2 - CH_2 - OH \xrightarrow[\Delta]{Conc.\ H_2SO_4}$

(B) $CH_3 - CH_2 - \underset{\displaystyle OH}{\underset{|}{CH}} - CH_3 \xrightarrow{1.\ TsCl/Pyridine}$

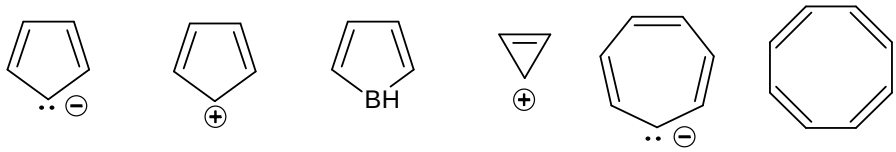
(C) $CH_3 - CH_2 - \underset{\displaystyle Cl}{\underset{|}{CH}} - CH_3 \xrightarrow[\Delta]{Alc.\ KOH}$

(D) $CH_3 - CH_2 - \underset{\displaystyle Cl}{\underset{|}{CH}} - CH_3 \xrightarrow[Bu^tOH/\Delta]{NaOBu^t}$

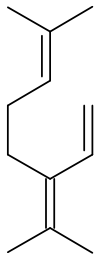
SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. Find out the number of species which are destabilized through conjugation?



47. Find out the number of different products obtained on the ozonolysis of the following compounds:

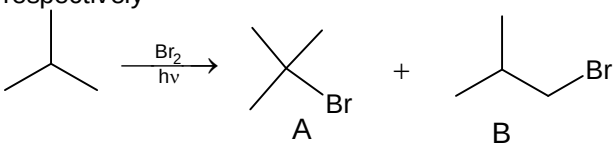


SECTION – C
(Numerical Answer Type)

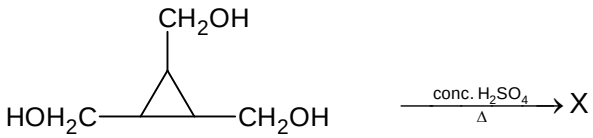
This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. Optically pure (S) (+) 2-bromo octane (α = 36°) reacts with aqueous NaOH in acetone to give optically pure (R) (–) 2 octanol, (α = - 10.3^o). With partially racemised bromo compound whose (α = 30°). Specific rotation of alcohol is – 6.0^o. What is the optical purity of partially racemised product?

49. The relative reactivity of 1°H, 2°H, & 3°H in bromination reaction if found to be 1 : 82 : 1600 respectively



What is the % yield of A?

50. 

If the molecular wt. of X is M, then $\frac{M}{5} = ?$

Mathematics

PART – III

SECTION – A

(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. The least value of n for which $(n-2)x^2 + 8x + (n+4) > \sin^{-1}(\sin 12) + \cos^{-1}(\cos 12)\ \forall\ x \in R$ ($n \in N$) is
(A) 4
(B) 5
(C) 6
(D) 7

52. In a triangle ABC, $\sum_{r=0}^{n} {}^nC_r a^r b^{n-r} \cos(rB-(n-r)A)$ is equal to (where a, b, c represents the sides and A, B, C represents the angles of ΔABC)
(A) a^n
(B) b^n
(C) c^n
(D) none of these

53. The radius of the circle touching the pair of lines $7x^2 - 18xy + 7y^2 = 0$ and the circle $x^2 + y^2 - 8x - 8y = 0$ and is contained in the given circle is
(A) $\sqrt{10}$
(B) 3
(C) $2\sqrt{2}$
(D) $\sqrt{7}$

54. A parabola having directrix $x + y + 2 = 0$ touches a line $2x + y - 5 = 0$ at (2, 1). Then the semi latus rectum of the parabola is
(A) $\frac{8}{\sqrt{2}}$
(B) $\frac{9}{\sqrt{2}}$
(C) $\frac{10}{\sqrt{2}}$
(D) $\frac{11}{\sqrt{2}}$

55. Number of points from which pair of perpendicular tangents can be drawn to hyperbola $x^2 \operatorname{cosec}^2 \alpha - y^2 \sec^2 \alpha = 1$, is $\alpha \in \left(0, \frac{\pi}{4}\right)$
(A) 0
(B) 1
(C) 2
(D) infinite

56. Let AB be the chord of contact of the point (5, –5) w.r.t the circle $x^2 + y^2 = 5$, then the locus of the orthocentre of ΔPAB where P is any point moving on the circle is
(A) $(x - 1)^2 + (y + 1)^2 = 5$
(B) $(x - 1)^2 + (y + 1)^2 = \frac{5}{2}$
(C) $(x + 1)^2 + (y - 1)^2 = 5$
(D) $(x + 1)^2 + (y - 1)^2 = \frac{5}{2}$

57. A parabola $y = ax^2 + bx + c$ crosses the x-axis at $(\alpha, 0)$, $(\beta, 0)$ both to the right of the origin. A circle also passes through these 2 points. The length of tangent from the origin to the circle is
(A) $\sqrt{\frac{bc}{a}}$
(B) ac^2
(C) $\frac{b}{a}$
(D) $\sqrt{\frac{c}{a}}$

58. If e_1 and e_2 are the roots of the equation $x^2 - ax + 2 = 0$ where e_1 and e_2 are the eccentricities of an ellipse and hyperbola respectively, then a belongs to
(A) $(3, \infty)$
(B) $(2, \infty)$
(C) $(1, \infty)$
(D) $(-\infty, 1) \cup (1, 2)$

59. If the locus of a point, whose chord of contact with respect to the circle $x^2 + y^2 = 4$ is a tangent to the curve $xy = 1$ is $xy = c^2$, then the value of c^2 is
(A) 2
(B) 4
(C) $\frac{1}{2}$
(D) $\frac{1}{4}$

60. Let $f(x) = \sin x + \cos x + \tan x + \sin^{-1} x + \tan^{-1} x + \cos^{-1} x$. If M and m are maximum and minimum values of f(x), then its sum is
(A) $\pi + 2 \cos 1$
(B) $\pi + 2 \sin 1$
(C) $\frac{\pi}{2} + 2\tan 1$
(D) $\pi + \tan 1 + \sin 1$

61. Consider an obtuse angled triangle with sides 8 cm, 15 cm and x cm (largest side being 15 cm). If x is an integer, then number of possible triangles are
(A) 4
(B) 6
(C) 7
(D) 5

62. In a right angled triangle, the smallest angle is θ. A triangle which is taking the reciprocal of sides is also a right angled triangle, then sin θ is equal to

(A) $\frac{\sqrt{5}+1}{4}$

(B) $\frac{\sqrt{5}-1}{2}$

(C) $\frac{\sqrt{5}-1}{4}$

(D) $\frac{\sqrt{2}-1}{2}$

63. If the equation of pair of straight lines passing through (1, 1), one making an angle θ with the positive direction of x-axis and other making the same angle with positive direction of y-axis is $x^2 - (a + 2)xy + y^2 + a(x + y - 1) = 0$, $a \neq -2$, then sin 2θ is

(A) $(a - 2)$

(B) $a + 2$

(C) $\frac{2}{a+2}$

(D) $\frac{2}{a}$

64. If the chords of the hyperbola $x^2 - y^2 = 16$ touches the parabola $y^2 = 16x$, then the locus of the middle points of these chords is a curve

(A) $y^2(x + 4) = x^3$

(B) $y^2(x - 4) = x^3$

(C) $y^2(x + 8) = 3x^3$

(D) $y^2(x - 8) = 2x^3$

65. If the number of radical axis of a system of circles is same as number of radical centres, then number of circles in that particular system is

(A) 4

(B) 6

(C) 10

(D) 5

66. The vertices of a ΔABC lie on a rectangular hyperbola such that the orthocentre of the triangle is (3, 2) and the asymptotes of the rectangular hyperbola are parallel to coordinate axes. If the two perpendicular tangents of the hyperbola intersect at (1, 1), then equation of the rectangular hyperbola is

(A) $xy = 2x + y - 2$

(B) $2xy = x + 2y + 5$

(C) $xy = x + y + 1$

(D) none of these

67. The locus of the centre of a variable circle touching two circles of radius r_1 and r_2 externally which also touch each other externally is a conic. The eccentricity of the conic, if $\frac{r_1}{r_2} = 3 + 2\sqrt{2}$ is

(A) 1

(B) $\sqrt{2}$

(C) $\frac{1}{2}$

(D) $2\sqrt{2}$

68. AB is any chord of the circle $x^2 + y^2 - 6x - 8y - 11 = 0$ which subtends an angle of $\frac{\pi}{2}$ at (1, 2). If locus of mid-point of AB is $x^2 + y^2 - 2ax - 2by - c = 0$, then a + b + c is
(A) 8
(B) 10
(C) 6
(D) 12

69. $\lim_{n\to\infty} \sum_{r=1}^{n} \cot^{-1}\left(\frac{r^3 - r + \frac{1}{r}}{2}\right)$ is equal to
(A) 0
(B) π
(C) $\frac{\pi}{2}$
(D) $\frac{\pi}{4}$

70. Let $f(\theta) = \frac{1}{1+(\cot\theta)^x}$, then the value of $2\sum_{\theta=1^\circ}^{89^\circ} f(\theta)$ is
(A) 44
(B) 89
(C) 65
(D) 178

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. The value of cot 76° cot 44° + cot 16° cot 44° – cot 76° cot 16° is

72. In a triangle ABC if $r_1 + r_3 + r = r_2$, then the value of $(\sec^2 A + \cos^2 B - \cot^2 C)$ is

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. Let ABC be a right angled triangle at C. If the inscribed circle touches the side AB at D and (AD)·(BD) = 11, then area of triangle is

74. If the reflection of the ellipse $\frac{(x-4)^2}{16} + \frac{(y-3)^2}{9} = 1$ in the line mirror x – y – 2 = 0 is $k_1x^2 + k_2y^2 - 160x - 36y + 292 = 0$, then $k_1 + k_2$ is equal to

75. A chord is drawn passing through P(2, 2) on the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ such that it intersects the ellipse at A and B, then the maximum value of PA·PB is